

## CO4\_AT3 – COMPARATIVE ANALYSIS: ANSWERS

### 1. Object Detection Performance Comparison

#### Given Data

| Model           | True Positives (TP) | False Positives (FP) | False Negatives (FN) |
|-----------------|---------------------|----------------------|----------------------|
| A (Viola-Jones) | 80                  | 20                   | 30                   |
| B (YOLO)        | 90                  | 15                   | 20                   |

#### (a) Precision and Recall Calculation

Formulas used: Precision =  $TP / (TP + FP)$     Recall =  $TP / (TP + FN)$

##### Model A (Viola-Jones)

- Precision =  $80 / (80 + 20) = 80 / 100 = 0.80 \rightarrow 80\%$
- Recall =  $80 / (80 + 30) = 80 / 110 = 0.7273 \rightarrow 72.73\%$

##### Model B (YOLO)

- Precision =  $90 / (90 + 15) = 90 / 105 = 0.8571 \rightarrow 85.71\%$
- Recall =  $90 / (90 + 20) = 90 / 110 = 0.8182 \rightarrow 81.82\%$

#### Summary Table

| Model           | Precision | Recall |
|-----------------|-----------|--------|
| A (Viola-Jones) | 80.00%    | 72.73% |
| B (YOLO)        | 85.71%    | 81.82% |

#### (b) Comparison and Evaluation

- Precision: Model B (85.71%) is higher than Model A (80.00%), meaning Model B produces fewer false positives relative to its total positive detections — it is more reliable when it flags an object as detected.
- Recall: Model B (81.82%) is higher than Model A (72.73%), meaning Model B misses fewer actual objects (fewer false negatives) — it detects a greater proportion of the true objects present.

Model B outperforms Model A on both precision and recall, so it dominates on every metric — there is no trade-off to weigh here.

## Justification

Model B (YOLO) provides better overall object detection performance. Since it has both higher precision (fewer false alarms) and higher recall (fewer missed detections) than Model A, it is unambiguously the stronger model for this dataset. This can also be confirmed by combining both metrics into the F1-score:

- $F1(\text{Model A}) = 2 \times (0.80 \times 0.7273) / (0.80 + 0.7273) = 2 \times 0.5818 / 1.5273 \approx 0.7619 \rightarrow 76.19\%$
- $F1(\text{Model B}) = 2 \times (0.8571 \times 0.8182) / (0.8571 + 0.8182) = 2 \times 0.7013 / 1.6753 \approx 0.8372 \rightarrow 83.72\%$

Model B's higher F1-score (83.72% vs. 76.19%) confirms it delivers a better overall balance of precision and recall, making it the superior object detection model.

## 2. Motion Estimation Error Analysis

### Given Data (Displacement Errors in Pixels)

| Method | Frame 1 | Frame 2 | Frame 3 |
|--------|---------|---------|---------|
| A      | 2       | 3       | 2       |
| B      | 3       | 5       | 4       |

#### (a) Average Displacement Error

Formula used: Average Error = (Sum of errors) / (Number of frames)

- Method A:  $(2 + 3 + 2) / 3 = 7 / 3 = 2.33$  pixels
- Method B:  $(3 + 5 + 4) / 3 = 12 / 3 = 4.00$  pixels

#### (b) Comparison

Method A has a lower average displacement error (2.33 pixels) compared to Method B (4.00 pixels). Method A is also more consistent, with errors ranging only from 2 to 3 pixels, whereas Method B's errors range more widely from 3 to 5 pixels, indicating greater variability and less reliability across frames.

Therefore, Method A provides more accurate motion estimation.

#### (c) Justification

A lower average displacement error means the estimated motion vectors are closer to the true motion between frames, i.e., less deviation from ground truth. Method A's average error of 2.33 pixels is 41.75% lower than Method B's 4.00 pixels, representing a substantial accuracy improvement. Additionally, Method A's smaller frame-to-frame variation (2, 3, 2) suggests more stable and predictable tracking performance, which is important for reliable motion estimation in real-world video sequences. For these reasons, Method A is the more accurate and dependable technique.

### 3. Depth Estimation Comparison

Given Data (in meters)

| Object | Actual Depth | Stereo Vision | SfM |
|--------|--------------|---------------|-----|
| 1      | 2.0          | 2.0           | 2.5 |
| 2      | 3.0          | 3.0           | 3.8 |
| 3      | 4.0          | 4.0           | 5.0 |

#### (a) Absolute Error for Each Object

Formula used: Absolute Error = |Estimated Depth – Actual Depth|

##### Stereo Vision

- Object 1:  $|2.0 - 2.0| = 0.0$  m
- Object 2:  $|3.0 - 3.0| = 0.0$  m
- Object 3:  $|4.0 - 4.0| = 0.0$  m

##### Structure-from-Motion (SfM)

- Object 1:  $|2.5 - 2.0| = 0.5$  m
- Object 2:  $|3.8 - 3.0| = 0.8$  m
- Object 3:  $|5.0 - 4.0| = 1.0$  m

#### Summary of Absolute Errors

| Object | Stereo Vision Error (m) | SfM Error (m) |
|--------|-------------------------|---------------|
| 1      | 0.0                     | 0.5           |
| 2      | 0.0                     | 0.8           |
| 3      | 0.0                     | 1.0           |

#### (b) Average Absolute Error

- Stereo Vision:  $(0.0 + 0.0 + 0.0) / 3 = 0.0 / 3 = 0.0$  m
- SfM:  $(0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = 0.7667$  m  $\approx 0.77$  m

#### (c) Comparison and Evaluation

Stereo Vision achieved a perfect match with the actual depths for all three objects, resulting in zero average absolute error. Structure-from-Motion (SfM), on the other hand, consistently overestimated the depth for every object, with the error increasing as the actual depth increased (0.5 m, 0.8 m, then 1.0 m), giving an average absolute error of approximately 0.77 m.

Stereo Vision provides significantly more accurate depth estimation than SfM for this dataset.

- Accuracy: Stereo Vision's 0.0 m average error vastly outperforms SfM's 0.77 m average error.
- Error trend: SfM's errors grow with distance, suggesting a systematic scale/drift issue common in monocular or motion-based depth estimation, which relies on triangulating features across frames and is more sensitive to accumulated errors, whereas Stereo Vision directly triangulates depth from two fixed, calibrated camera viewpoints, giving it higher geometric precision.

Therefore, for applications requiring precise depth measurement, Stereo Vision is the clearly preferable method based on this data.